

Engineering Physics — Formula & Derivation Handbook

Quick Reference for T1, T2 & T3 Examinations • Solved Derivations

1. Wave Optics & Interference

- Interference in Thin Films: $2\mu t \cos(r) = n\lambda$ (Constructive for reflection with phase shift)
- Newton's Rings Diameter: $D_n^2 = 4nR\lambda$ (Bright rings: $D_n^2 = 2(2n-1)R\lambda$)
- Diffraction Grating: $(a + b) \sin \theta = n\lambda$ (Grating element = $a + b$)

2. Quantum Mechanics Essentials

- de Broglie Wavelength: $\lambda = h / p = h / (mv) = h / \sqrt{2mE}$
- Heisenberg Uncertainty Principle: $\Delta x \cdot \Delta p \geq \hbar / 2$
- Time-Independent Schrödinger Equation: $-\left(\hbar^2 / 2m\right) \nabla^2 \psi + V\psi = E\psi$